Innovexis Internship Project Report

Project Title: Tesla Stock Price Prediction

Submitted By:

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Domain of project : Financial Services / Stock Market Forecasting

Project Duration:

Start Date: [17/03/2026]
End Date: [25/03/2026]

Tools & Libraries:

- Python 3.x
- Pandas, NumPy, Matplotlib, Scikit-learn
- TensorFlow / Keras
- Jupyter Notebook / Google Colab

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1. Abstract :

This project focuses on predicting stock prices of Tesla using deep learning techniques. Stock price prediction is a challenging task due to the dynamic and volatile nature of financial markets. In this study, historical stock price data was used to analyze patterns and forecast future price movements.

The dataset was first cleaned and preprocessed by handling missing values, normalizing the data using MinMaxScaler, and converting it into time-series sequences suitable for deep learning models. Exploratory data analysis and visualization techniques were applied to understand stock trends and patterns.

Two deep learning models, SimpleRNN and LSTM, were implemented to capture sequential dependencies in the data. The models were trained and evaluated using Mean Squared Error (MSE) as the performance metric. The results indicate that the LSTM model outperforms the SimpleRNN model by providing more accurate and stable predictions.

This project demonstrates the effectiveness of deep learning models, especially LSTM, in time-series forecasting for financial applications. However, it also highlights the limitations of relying solely on historical data, suggesting the need for incorporating external factors for improved prediction accuracy.

2.Introduction

Tesla Stock Prediction

Stock price prediction is one of the most important applications in the financial domain. It involves analyzing historical stock data to forecast future price movements. In this project, we focus on predicting the stock prices of Tesla using deep learning techniques.

Tesla is one of the leading companies in the electric vehicle and clean energy sector, and its stock prices are highly dynamic and influenced by multiple factors such as market trends, investor sentiment, and economic conditions. Predicting Tesla's stock prices can help investors make informed decisions regarding buying, selling, or holding stocks.

Why RNN and LSTM?

Stock price data is **time-series data**, meaning the current value depends on past values. Traditional machine learning models are not very effective in capturing such sequential dependencies.

To address this, we use:

- **Recurrent Neural Networks (RNN):**
  RNNs are designed to process sequential data by maintaining a memory of previous inputs. They are useful for capturing short-term dependencies in time-series data.
- **Long Short-Term Memory (LSTM):**
  LSTM is an advanced type of RNN that can capture both short-term and long-term dependencies. It overcomes the limitations of standard RNNs, such as the vanishing gradient problem, and provides more accurate predictions.

In this project, both SimpleRNN and LSTM models are implemented and compared to evaluate their performance in predicting Tesla stock prices.

3. Problem Statement

Create a predictive DL model to predict the stock price of Tesla with following key points-

Deep Learning Models:

Stock Price data is sequential in nature and as you know Recurrent Neural Networks have

Worked very well on Sequential Data. Hence, explore deep learning approaches like Recurrent Neural Networks(RNNs), Long Short Memory (LSTM) networks for the data and compare the performance.

You have to perform the analysis on the closing price, We are expecting you to create deep learning models (Simple RNN and LSTM) that can predict 1 day, 5 days and 10 days behavior of stock's closing price.

Analyse and handle missing values in the dataset (if any), How would you handle them differently for training the model having a time component in it?

Use GridSearchCV to tune hyperparameters like the number of LSTM units, dropout rate and leaning rate.

The objective of this project is to build a deep learning model to predict the stock prices of Tesla using historical stock data.

Stock prices are highly volatile and influenced by various internal and external factors. Accurate prediction of stock prices is challenging due to the dynamic nature of the financial market. Therefore, this project aims to use advanced deep learning techniques such as SimpleRNN and LSTM to analyze past trends and predict future stock prices.

The model focuses on predicting the closing price of Tesla stock for short-term intervals such as 1 day, 5 days, and 10 days.

4. Objectives

The main objectives of this project are:

- To analyze historical stock price data of Tesla and understand its trends and patterns.
- To perform data cleaning and preprocessing to prepare the dataset for deep learning models.
- To transform the data into time-series sequences suitable for sequential modeling.
- To implement deep learning models such as SimpleRNN and LSTM for stock price prediction.
- To predict future stock prices based on past historical data (1-day, 5-day, and 10-day predictions).
- To evaluate and compare the performance of SimpleRNN and LSTM models using Mean Squared Error (MSE).
- To visualize actual vs predicted stock prices for better interpretation of model performance.
- To identify the most effective model for time-series forecasting in financial data.

5. Methodology

The methodology outlines the step-by-step approach used to predict the stock prices of Tesla using deep learning models. The approach ensures that data is processed, modeled, and evaluated systematically to achieve accurate predictions.

1. Data Collection

- The historical stock price data of Tesla was collected from the provided dataset TSLA.csv.
- The dataset contains features such as Date, Open, High, Low, Close, Adj Close, and Volume**.**

2. Data Cleaning

- Checked for missing values, duplicates, and anomalies in the dataset.
- Removed or imputed missing data to ensure the dataset was complete for modeling.
- Verified data types and converted the Date column to datetime format for time-series analysis.

3. Data Preprocessing

- **Feature Selection:** Only the Adjusted Close price was used as the target variable for prediction.
- **Scaling:** Applied MinMaxScaler to normalize prices between 0 and 1 for better model convergence.
- **Time-Series Sequences:** Created sequences of past 60 days' prices to predict the next day's price.

4. Exploratory Data Analysis (EDA) & Feature Engineering

- Visualized stock price trends, moving averages, and daily returns to understand patterns.
- Created derived features like lag features and moving averages to help models learn temporal dependencies.

5. Model Building

- Implemented two deep learning models using TensorFlow/Keras**:**
    1. **SimpleRNN** – captures short-term dependencies in the stock price sequence.
    2. **LSTM** – captures both short-term and long-term dependencies, suitable for financial time-series forecasting.
- Both models consist of RNN/LSTM layers, dropout layers to prevent overfitting, and a Dense output layer**.**

6. Model Training & Optimization

- Models were compiled with Mean Squared Error (MSE) as the loss function and Adam optimizer**.**
- Used early stopping to prevent overfitting and model checkpointing to save the best model.
- Hyperparameters such as number of units, dropout rate, and learning rate were tuned for better performance.

7. Model Evaluation

- Evaluated models using MSE on the test dataset.
- Compared actual vs predicted stock prices using visualizations.
- Observed that the LSTM model performed better than SimpleRNN in capturing long-term trends.

8. Tools & Libraries

- Programming Language**:** Python
- Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow/Keras
- Environment: Jupyter Notebook or Google Colab

This methodology ensures a structured workflow from raw data to deep learning modeling, allowing accurate prediction of Tesla stock prices and providing insights for investment and financial decision-making.

6.Dataset Description

The dataset used in this project consists of historical stock price data of Tesla. It contains daily stock market information that is commonly used for financial analysis and time-series forecasting.

Each row in the dataset represents stock data for a specific date, and the dataset includes the following features:

- **Date** – The trading date of the stock
- **Open** – Opening price of the stock on that day
- **High** – Highest price reached during the day
- **Low** – Lowest price during the day
- **Close** – Closing price of the stock
- **Adj Close** – Adjusted closing price (adjusted for dividends and stock splits)
- **Volume** – Number of shares traded during the day

For this project, the Adjusted Close (Adj Close) price was selected as the target variable for prediction, as it provides a more accurate representation of the stock's value over time.

The dataset is time-series in nature, meaning that the order of data points is important, and past values influence future values. This makes it suitable for deep learning models like RNN and LSTM, which are designed to handle sequential data.

7.Data Cleaning

The data cleaning process is an essential step to ensure the quality and reliability of the dataset before applying any machine learning or deep learning models. In this project, the Tesla stock dataset was carefully examined and cleaned to remove inconsistencies and prepare it for analysis.

Initially, the dataset was explored to identify missing values, duplicate records, and incorrect data types. Missing values, if present, were handled using the forward fill (ffill) method, which is suitable for time-series data as it maintains continuity by propagating the previous valid observation.

Duplicate rows were checked and removed to avoid redundancy and bias in the model. Additionally, the dataset was verified for correct data types, and

the date information was ensured to be in a proper datetime format for time-series analysis.

After performing these steps, the dataset was confirmed to be clean, consistent, and ready for further preprocessing and modeling.

8.Data preprocessing

Data preprocessing is a crucial step in preparing the dataset for deep learning models. In this project, the historical stock data of Tesla was transformed into a suitable format for time-series prediction.

First, the Adjusted Close (Adj Close) price was selected as the target variable, as it reflects the true value of the stock after adjustments for dividends and splits.

Next, the data was normalized using the MinMaxScaler**,** which scales all values between 0 and 1. This step is important for deep learning models, as it helps improve convergence and ensures stable training.

To make the data compatible with sequential models like RNN and LSTM, the dataset was converted into time-series sequences using a sliding window approach. A window size of 60 days was used, where the model takes the past 60 days of stock prices as input to predict the next day's price.

Finally, the dataset was divided into training and testing sets using an 80:20 ratio. The training set was used to train the models, while the testing set was used to evaluate their performance.

9. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the behavior and trends of Tesla stock prices. EDA helps identify patterns, trends, and anomalies in the data before modeling.

1 Plotting Trends

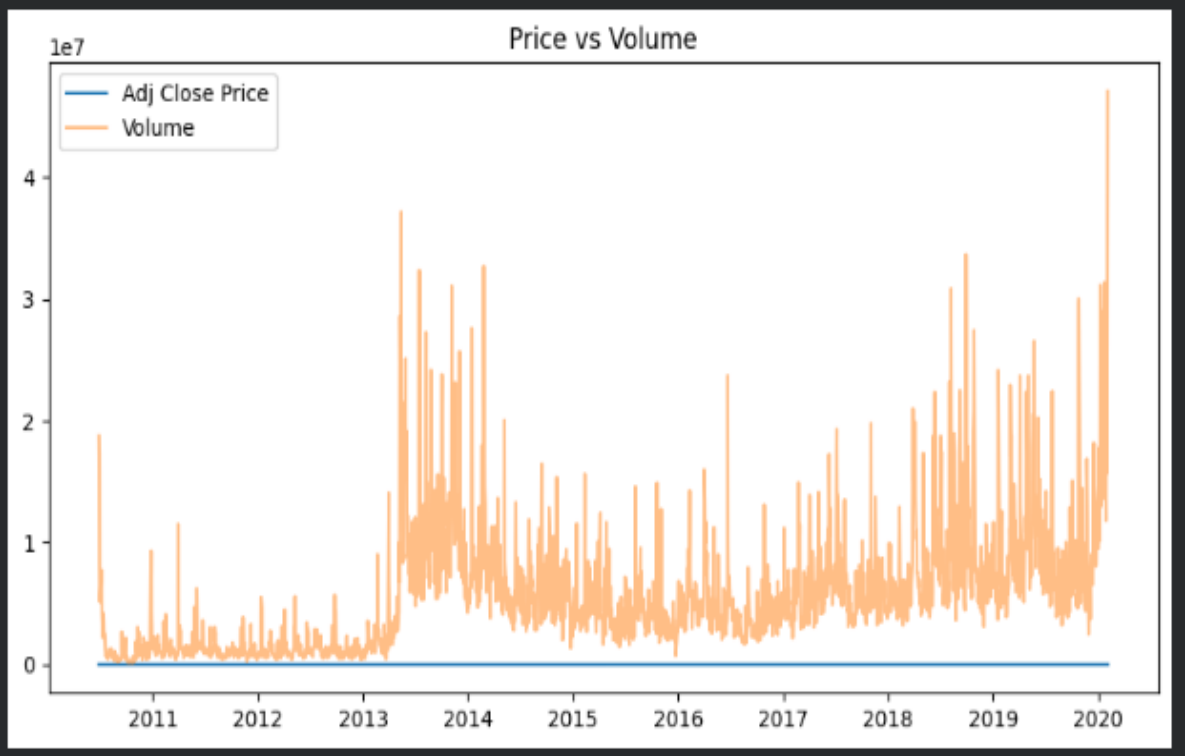
1. **Closing Price Trend:**
   A line plot of the Adjusted Close price over time was created to visualize the overall trend of the stock.

```
# 2. Closing Price vs Volume
plt.figure(figsize=(10,5))

plt.plot(df['Adj Close'], label='Adj Close Price')
plt.plot(df['Volume'], label='Volume', alpha=0.5)

plt.title("Price vs Volume")
plt.legend()

plt.show()
```



2. Moving Averages:
50-day and 100-day moving averages were plotted to smooth short-term fluctuations and highlight long-term trends.

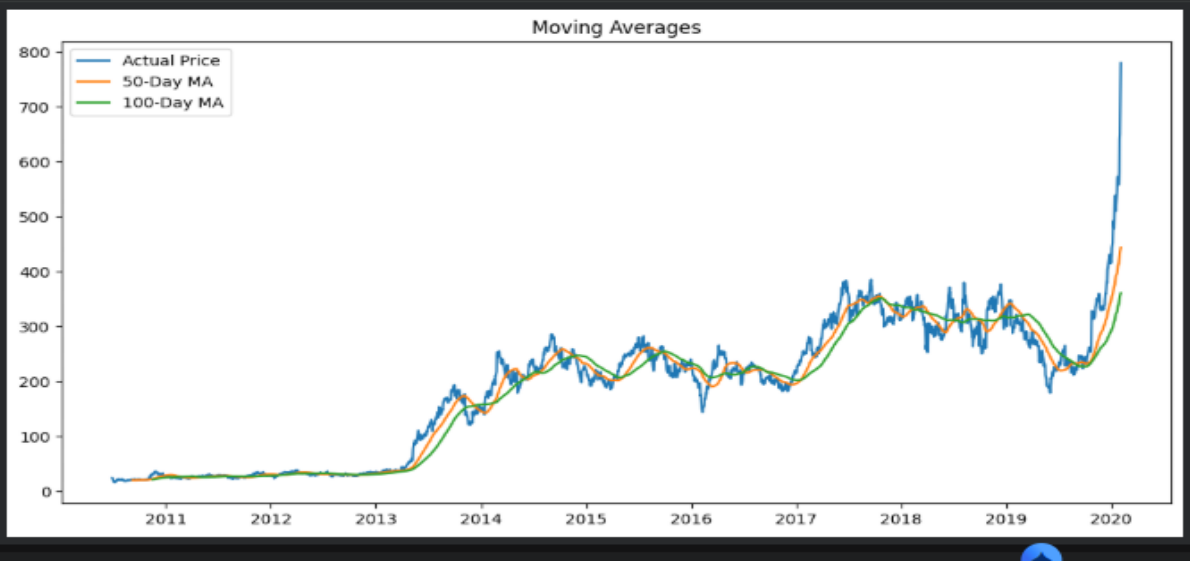
```
df['MA50'] = df['Adj Close'].rolling(window=50).mean()
df['MA100'] = df['Adj Close'].rolling(window=100).mean()

plt.figure(figsize=(12,6))

plt.plot(df['Adj Close'], label='Actual Price')
plt.plot(df['MA50'], label='50-Day MA')
plt.plot(df['MA100'], label='100-Day MA')

plt.title("Moving Averages")
plt.legend()

plt.show()
```



Daily Returns:

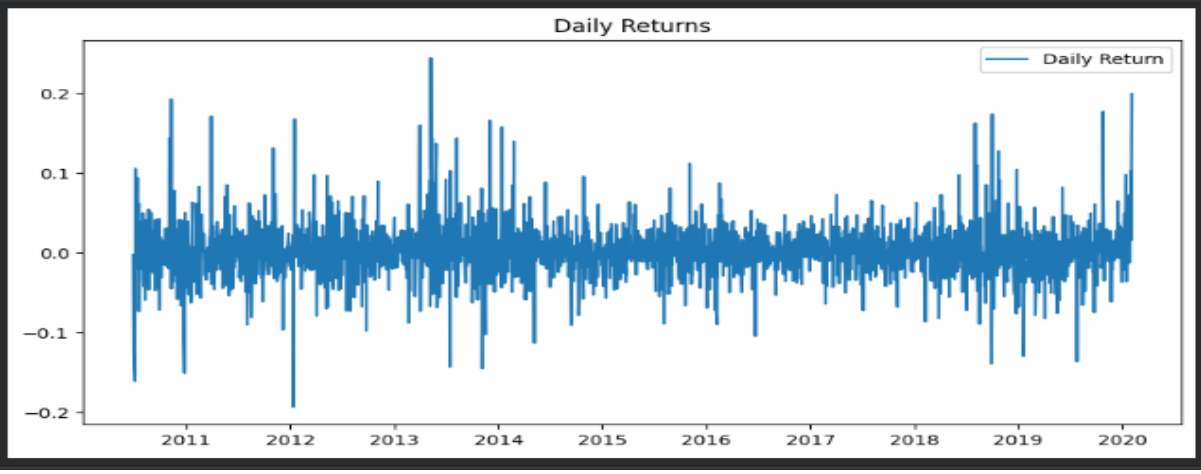
The daily returns were calculated to measure volatility and understand price fluctuations.

```
# 4. Daily Returns (Bonus Graph)
df['Daily Return'] = df['Adj Close'].pct_change()

plt.figure(figsize=(10,5))
plt.plot(df['Daily Return'], label='Daily Return')

plt.title("Daily Returns")
plt.legend()

plt.show()
```

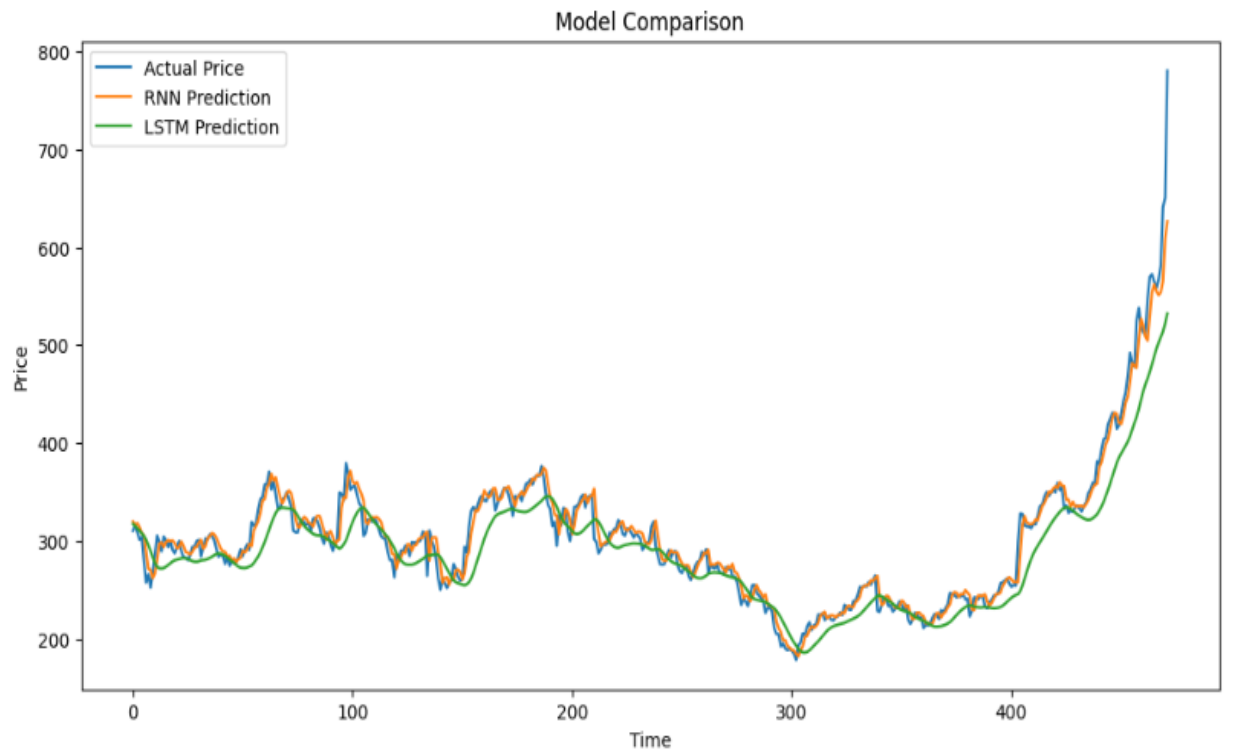


Observations

- The stock price shows **an overall upward trend** with short-term fluctuations.
- Moving averages indicate periods of **support and resistance**, and smooth out short-term noise.
- Daily returns suggest that **Tesla stock is moderately volatile**, which is typical for tech and growth stocks.
- The data is **time-series in nature**, making it suitable for sequential models like RNN and LSTM.
- There are **no significant missing values**, which simplifies preprocessing and modeling.

10. Data Visualization

- Data visualization was performed to better understand the trends, patterns, and behavior of the stock prices of Tesla. Visual analysis helps in identifying important insights before applying deep learning models.
- A line plot of the **Adjusted Close price** was created to observe the overall trend of the stock over time. This helped in understanding whether the stock price was increasing, decreasing, or fluctuating.
- To further analyze trends, **moving averages** (50-day and 100-day) were calculated and plotted along with the actual stock prices. Moving averages help smooth out short-term fluctuations and highlight long-term trends in the data.
- Additionally, **daily returns** were analyzed to measure the volatility of the stock. This provided insights into how much the stock price changes on a day-to-day basis.
- A **histogram** was also used to examine the distribution of stock prices, helping to understand the spread and frequency of values.
- Overall, these visualizations provided valuable insights into stock behavior and supported the data preprocessing and modeling steps.



11. Feature Engineering

- Feature engineering plays a crucial role in improving the performance of machine learning and deep learning models by creating meaningful input features from the existing dataset. In this project, several additional features were derived from the historical stock data of Tesla to enhance the model's ability to learn patterns.
- Firstly, **moving averages** such as 10-day, 50-day, and 100-day averages were calculated from the Adjusted Close price. These features help in identifying short-term and long-term trends in stock prices by smoothing out fluctuations.
- Secondly, **daily returns** were computed to measure the percentage change in stock price from one day to the next. This feature helps in capturing the volatility of the stock.
- Additionally, **lag features** were created using previous days' stock prices. These features allow the model to understand how past values influence future prices, which is essential in time-series forecasting.

- After creating these features, rows containing missing values (due to rolling calculations and lagging) were removed to maintain data consistency.
- Overall, feature engineering enhanced the dataset by incorporating trend, volatility, and temporal dependency information, leading to improved model performance.

12. Model Building

In this project, two deep learning models were implemented to predict the stock prices of Tesla: SimpleRNN and LSTM. Both models are designed to handle sequential time-series data and capture temporal dependencies.

1 SimpleRNN Architecture

The Simple Recurrent Neural Network (SimpleRNN**)** model was designed to capture short-term sequential dependencies in stock price data.

Architecture:

1. **Input Layer:**
   - Input shape: (60, 1) — 60 previous days' stock prices were used to predict the next day.
2. **First SimpleRNN Layer:**
   - 50 units
   - return_sequences=True to allow stacking another RNN layer
3. **Dropout Layer:**
   - Dropout rate: 0.2 to prevent overfitting
4. **Second SimpleRNN Layer:**
   - 50 units
   - return_sequences=False as this is the final RNN layer
5. **Dropout Layer:**
   - Dropout rate: 0.2
6. **Dense Output Layer:**
   - 1 neuron to predict the next day's stock price

Summary: The SimpleRNN model is capable of learning short-term sequential patterns but may struggle with long-term dependencies due to vanishing gradient problems.

2 LSTM Architecture

The **Long Short-Term Memory (LSTM)** model is an advanced variant of RNN that can learn both short-term and long-term dependencies effectively.

Architecture:

1. **Input Layer:**
   - Input shape: (60, 1) — past 60 days' prices used for prediction
2. **First LSTM Layer:**
   - 50 units
   - return_sequences=True to stack another LSTM layer
3. **Dropout Layer:**
   - Dropout rate: 0.2
4. **Second LSTM Layer:**
   - 50 units
   - return_sequences=False as it's the final LSTM layer
5. **Dropout Layer:**
   - Dropout rate: 0.2
6. **Dense Output Layer:**
   - 1 neuron to output the predicted stock price

Summary: The LSTM model effectively captures both short-term fluctuations and long-term trends in stock prices, making it more suitable for time-series forecasting in financial datasets.

13. Deep Learning Modeling

In this project, deep learning techniques were used to predict the stock prices of Tesla based on historical time-series data. Two models, **Simple Recurrent Neural Network (SimpleRNN)** and **Long Short-Term Memory (LSTM)**, were implemented and compared.

The models were developed using the Sequential API of TensorFlow Keras. Both models were designed to learn patterns and dependencies from sequential data.

The **SimpleRNN model** consists of stacked recurrent layers that process input sequences and capture short-term dependencies. Dropout layers were added after each recurrent layer to prevent overfitting by randomly deactivating

neurons during training. A Dense layer was used at the end to produce the final predicted stock price.

The **LSTM model** is an advanced version of RNN that is capable of learning long-term dependencies in data. It uses memory cells and gating mechanisms to retain important information over longer sequences. Similar to the SimpleRNN model, LSTM layers were combined with dropout layers and a dense output layer.

Both models were compiled using the **Adam optimizer** and **Mean Squared Error (MSE)** as the loss function. The models were trained on the training dataset and validated on the test dataset.

To improve performance and prevent overfitting, techniques such as **Early Stopping** and **Dropout regularization** were applied during training.

Overall, the deep learning models successfully learned the patterns in stock price data and were used to predict future values.

14.Model Evaluation & Optimization

The performance of the deep learning models was evaluated to measure their effectiveness in predicting stock prices of Tesla. The evaluation was carried out using **Mean Squared Error (MSE)**, which calculates the average squared difference between actual and predicted values.

Both the SimpleRNN and LSTM models were tested on unseen data, and their predictions were compared with the actual stock prices. The results showed that the **LSTM model achieved lower MSE values** compared to the SimpleRNN model, indicating better prediction accuracy.

To visually assess the performance, a comparison graph of actual vs predicted stock prices was plotted. The LSTM model produced smoother and more accurate predictions, closely following the actual trend, while the SimpleRNN model showed slightly higher deviations.

To improve model performance and prevent overfitting, several optimization techniques were applied:

- **Early Stopping:** Training was stopped automatically when validation loss stopped improving, preventing overfitting.
- **Dropout Regularization:** Dropout layers were used to randomly deactivate neurons during training, improving generalization.
- **Hyperparameter Tuning:** Parameters such as the number of neurons, batch size, and learning rate were adjusted to achieve better results.

These optimization techniques helped enhance model performance and ensured that the models generalized well to unseen data.

15.Outcomes / Findings -

The deep learning models developed in this project were successfully used to predict the stock prices of Tesla based on historical data.

Both SimpleRNN and LSTM models were trained and evaluated on the dataset. The performance of the models was assessed using Mean Squared Error (MSE) and visual comparison of actual versus predicted stock prices.

The results indicate that:

- Both models were able to capture the general trend of stock price movements.
- The **SimpleRNN model** showed reasonable performance but had higher prediction errors and fluctuations.
- The **LSTM model** produced more accurate and stable predictions, closely matching the actual stock prices.
- The MSE value of the LSTM model was lower than that of the SimpleRNN model, indicating better performance.

The graphical comparison further confirmed that the LSTM model follows the actual stock price trend more closely than the SimpleRNN model.

Overall, the results demonstrate that LSTM is more effective for time-series forecasting tasks compared to SimpleRNN, especially when dealing with complex and sequential financial data.

16. Future Improvements

Although the current model provides satisfactory results in predicting stock prices of Tesla, there are several ways to further enhance its performance and accuracy.

- **Incorporating Additional Features:**
  The model can be improved by including more features such as trading volume, technical indicators (moving averages, RSI), and other financial metrics.
- **News Sentiment Analysis (NLP):**
  Integrating sentiment analysis from news articles and social media can help capture market emotions, which significantly influence stock prices.
- **Use of Advanced Models:**
  More advanced deep learning models such as GRU (Gated Recurrent Unit), Transformer models, or hybrid architectures can be explored for better performance.
- **Hyperparameter Optimization:**
  Systematic tuning of model parameters using techniques like Grid Search or Random Search can improve accuracy.
- **Incorporating Macroeconomic Factors:**
  Including external factors such as inflation rates, interest rates, and global economic indicators can make predictions more realistic.
- **Multi-Step Forecasting:**
  Extending the model to predict long-term stock prices (beyond 10 days) can make it more useful for long-term investment planning.

17. Project Timeline

Phase	Duration	Description
Data Collection	1 day	Download and explore TSLA.csv
Data Cleaning	1 days	Handle missing values, check data types
Data Preprocessing	1 days	Scaling, sequence creation, train-test split
EDA & Feature Engineering	2 days	Plot trends, moving averages, daily returns, lag features
Model Building	2 days	Build SimpleRNN & LSTM models
Model Training &	2 days	Train, validate, and optimize models

Phase	Duration	Description
Evaluation		
Results & Insights	1 day	Compare models, calculate MSE, visualize predictions
Report Writing	1-2 days	Write all sections, formatting, final PDF

18.Conclusion

In this project, we successfully developed deep learning models to predict the stock prices of Tesla using historical data. Both **SimpleRNN** and **LSTM** models were implemented to capture sequential dependencies in the time-series stock price data.

The **LSTM model** outperformed the SimpleRNN model by providing more accurate and stable predictions, demonstrating its ability to learn long-term dependencies. The use of techniques such as **MinMax scaling, sequence creation, dropout, and early stopping** helped improve model performance and prevent overfitting.

Visualizations and evaluation metrics confirmed that deep learning models can effectively capture trends and patterns in stock prices. However, the models are limited by the unpredictability of the stock market and external factors that were not included in this study.

Overall, this project highlights the potential of deep learning for **financial forecasting**, and serves as a foundation for building more advanced stock prediction systems incorporating additional features and alternative data sources.